



DATA SCIENCE ROADMAP (2026)



YOUR STEP-BY-STEP PATH TO BECOME A DATA SCIENTIST



High Demand
Career



Great Salary
Potential



Solve Real-World
Problems



Work in AI &
Future Tech

1 PYTHON FUNDAMENTALS

- Learn Python Basics
- Data Types, Loops, Functions
- OOP, Modules, Packages
- File Handling, Exceptions
- Practice Coding

Tools / Libraries



2 MATHEMATICS & STATISTICS

- Linear Algebra
- Calculus
- Probability & Statistics
- Descriptive Statistics
- Inferential Statistics
- Distributions

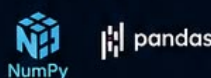
Key Topics



3 DATA HANDLING & ANALYSIS

- NumPy for Arrays
- Pandas for Data Manipulation
- Data Cleaning
- Data Wrangling
- Exploratory Data Analysis

Tools / Libraries



4 DATA VISUALIZATION

- Data Visualization Basics
- Matplotlib & Seaborn
- Charts: Line, Bar, Pie, Histogram, Boxplot
- Advanced Visualizations
- Storytelling with Data

Tools / Libraries



5 SQL & DATABASES

- SQL Basics
- SELECT, WHERE, GROUP BY
- JOINS, Subqueries
- Window Functions
- Data Modeling Basics
- Practice Queries

Tools



6 MACHINE LEARNING FUNDAMENTALS

- Supervised vs Unsupervised Learning
- Regression Algorithms
- Classification Algorithms
- Model Evaluation
- Cross Validation

Algorithms



7 ADVANCED MACHINE LEARNING

- Ensemble Methods
- Random Forest
- Gradient Boosting
- XGBoost
- Hyperparameter Tuning
- Handling Imbalanced Data

Tools / Libraries



8 DEEP LEARNING FUNDAMENTALS

- Neural Networks Basics
- Activation Functions
- Loss Functions
- Optimizers
- Training vs Validation
- Overfitting & Regularization

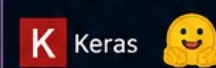
Tools / Libraries



9 DEEP LEARNING ADVANCED

- CNN (Computer Vision)
- RNN, LSTM (Sequences)
- Transfer Learning
- Attention Mechanism
- Model Fine-tuning
- NLP Basics

Tools / Libraries



10 DATA SCIENCE PROJECTS

- Work on End-to-End Projects
- Use Real-world Datasets
- EDA + Modeling + Deployment
- Build Portfolio
- Share on GitHub & Kaggle

Project Ideas



11 FEATURE ENGINEERING

- Feature Creation
- Feature Scaling
- Encoding Categorical Data
- Feature Selection
- Dimensionality Reduction (PCA)

Tools / Libraries



12 DATA STORYTELLING & COMMUNICATION

- Present Insights Clearly
- Create Dashboards
- Data Storytelling Principles
- Business Understanding
- Improve Communication

Tools



13 BIG DATA FUNDAMENTALS

- Big Data Concepts
- Hadoop Ecosystem
- Spark Basics
- Working with Large Datasets
- Data Pipelines

Tools



14 DEPLOYMENT & MLOPS

- Model Deployment Basics
- Flask / FastAPI
- Docker Basics
- CI/CD for ML Models
- MLflow / Monitoring
- Streamlit / Gradio Apps

Tools



15 CONTINUOUS LEARNING

- Stay Updated
- Read Research Papers
- Follow Communities
- Learn New Tools
- Build & Share Projects

Platforms



LEARNING PATH FLOW



ESSENTIAL SOFT SKILLS

- ✓ Problem Solving
- ✓ Critical Thinking
- ✓ Communication
- ✓ Business Understanding
- ✓ Curiosity & Consistency



RESOURCES TO FOLLOW

- Kaggle
- Towards Data Science
- Analytics Vidhya
- Medium
- ArXiv Papers
- Data Science Podcasts



POPULAR CAREER ROLES



REMEMBER

Consistency + Practice + Projects = Success

Keep Building. Keep Learning. Keep Growing!



★ THE FUTURE IS DATA. THE FUTURE IS YOU.

START TODAY, BECOME IRREPLACEABLE TOMORROW.